



EXPERIMENT NO-6

**Study and construction of full Adder using fundamental
(basic) logic gates.**



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Experiment No – 6, Study and construct a Full Adder using Fundamental (basic) digital logic gates.

Theory:

A full adder is a digital circuit that performs addition on three binary inputs and produces two binary outputs. It's a fundamental building block in digital electronics and is used to add binary numbers. The three inputs to a full adder are typically labelled A, B, and C_{in} (carry-in), and the outputs are usually **Sum** (the sum of A, B, and C_{in}) and C_{out} (carry-out, which represents any carry generated by the addition).

C_{in}	B	A	Sum (S)	Carry (C_{out})
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Table 6.1, The truth table for a full adder.

The Sum output gives the result of adding A, B, and C_{in} , while the C_{out} output indicates if there's a carry generated, which is useful for chaining multiple full adders together to add multi-bit numbers.

Full adders can be used in combination to create adders for multi-bit binary numbers. A 4-bit binary adder, for example, would use four full adders chained together, with the C_{out} from one full adder connected to the C_{in} of the next full adder, and so on. This is how computers perform binary addition.

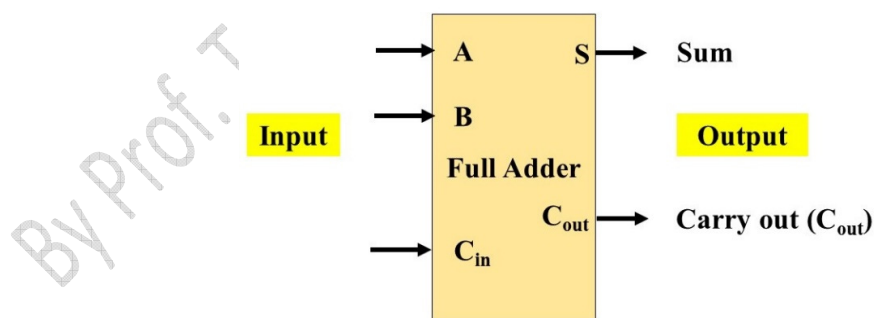


Figure (6.2), block diagram of full adder.

The SUM and CARRY outputs of a full adder can be expressed using Boolean logic. The Boolean expressions for SUM (S) and CARRY-OUT (C_{out}) in terms of the inputs A, B, and C_{in} are as follows:

SUM (S) expression:

$$S = \overline{C_{in}} \cdot \overline{B} \cdot A + \overline{C_{in}} \cdot B \cdot \overline{A} + C_{in} \cdot \overline{B} \cdot \overline{A} + C_{in} \cdot B \cdot A \dots\dots\dots (1)$$

Rearranging the terms, we can write the above expression as;

$$S = A \oplus B \oplus C_{in} \dots\dots\dots (2)$$

Here, $\oplus$ represents the XOR (exclusive OR) operation. The SUM is obtained by XORing the inputs A, B, and C_{in} .

Similarly, we can write the Boolean expression for the Carry also as given below;

$$C_{out} = \overline{C_{in}} \cdot B \cdot A + C_{in} \cdot \overline{B} \cdot A + C_{in} \cdot B \cdot \overline{A} + C_{in} \cdot B \cdot A \dots\dots\dots (3)$$

Simplifying, the above expression we can write it as follows;

$$C_{out} = C_{in} \cdot A + C_{in} \cdot B + B \cdot A \dots\dots\dots (4)$$

These Boolean expressions for the SUM and CARRY-OUT outputs of a full adder are essential in understanding how a full adder works and can be used to implement the full adder circuit using logic gates or in digital design.

Semiconductors and Instruments required

1. DC +5V regulated power supply.
2. Wish/Bread board.
3. Jumper wires.
4. Digital voltmeter.
5. Quad 2-input AND gate (7408)
6. Quad 2-input OR gate (7432)
7. Quad 2-input XOR (7486)

Implementation and Circuit diagram

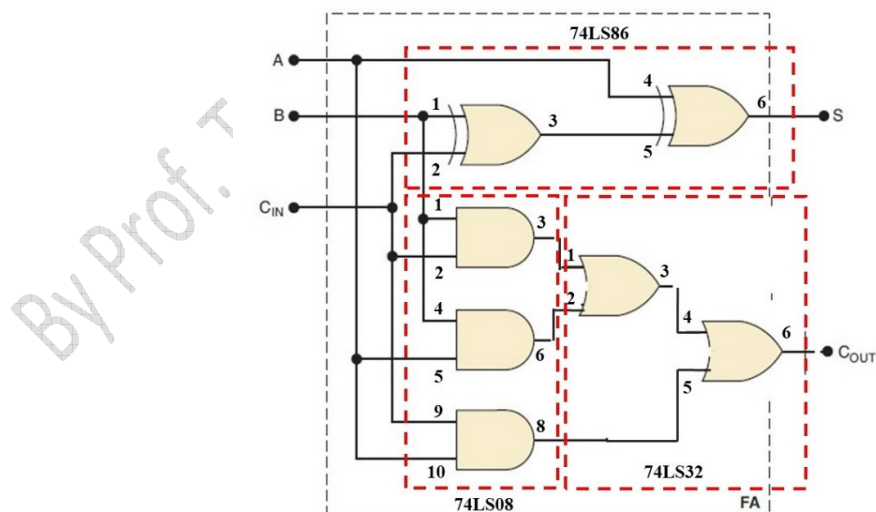


Figure (6.3), Implementation circuit diagram using fundamental / basic and other logic gate.

Result/Observation

Truth table for the full adder using fundamental / basic and other logic gates.

Logical Input			Logical output		Output voltage	
C	B	A	Sum (D)	Carry (Cy)	Sum (D)	Carry (Cy)
0	0	0	0	0		
0	0	1	1	0		
0	1	0	1	0		
0	1	1	0	1		
1	0	0	1	0		
1	0	1	0	1		
1	1	0	0	1		
1	1	1	1	1		

Table 6.2, Experimental observation.

Discussion

1. The power supply used must be regulated to +5V dc.
2. The jumpers used on the bread board must be short.
3. All the IC used must be properly seated on the bread board along with proper connection with regulated power supply.
4. The inputs to be connected to logic - 0 must not be left floating. For logic-1 and logic-0 we need to connect the inputs either to high voltage i.e. **+5V** (V_{CC}) or to the ground potential.
5. Outputs can be visualized by connecting LED. Those must be connected via current limiting resistor.
6. In this circuit we are implementing the sum part of the logic circuit using Quad 2-input XOR IC 74LS86. We can also implement the same using basic or fundamental gates also. The expression (1) can be implemented using basic gates. But here we have chosen to implement the logic expression (2).